



ABHISHEK KUMAR

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Research Summary

Independent interdisciplinary research focused on ambiguity-aware computational systems, multi-state computation models, Q-State dynamics, structured sparsity, graph-based computational frameworks, simulation methodologies, and computational evaluation systems.

Research Areas

- Ambiguity-Aware Multi-State Computation
- Q-State Dynamics
- Sparsity Dynamics
- State-Aware Graph Systems
- Computational Modeling and Simulation
- Artificial Intelligence and Computational Systems
- Infrastructure-Oriented Computational Research

Research Experience Highlights

- Computational Systems Research
- Graph-Based Evaluation Models
- State-Aware Systems
- Simulation & Experimental Studies
- Infrastructure-Oriented Research
- AI-Assisted Computational Research
- Responsible Computational Evaluation

Publications & Research Records

- Research Paper (2026) — Published / Public Record
- Figshare DOI — Available
- PhilArchive — Indexed
- PhilPapers — Indexed
- Scientific Research Journal Submission (2026) — Submitted
- Digital Commons Research Submission — Submitted

Simulation & Experimental Studies

- State Transition Analysis
- Q-State Evaluation Models
- Structured Sparsity Studies
- Graph Traversal Simulations
- Boundary Condition Experiments
- Computational Validation Studies

Patent Activities

US Patent Filed (USPTO)
Application No. 64/024,116 (2026)

Research Domains

- Artificial Intelligence
- Computer Science & Systems
- Numerical Methods
- Distributed Systems

Education Summary

- Computational Research
- Computational Engineering
- Computer Science & Logic
- Mathematics & Computational Reasoning
- Information Systems
- Business Systems

Academic Reference

Professor Virgilio E. Gonzalez
Associate Chair, Electrical & Computer Engineering
Professor of Practice, Electrical & Computer Engineering
Computational Science
The University of Texas at El Paso (UTEP)

Academic reference provided during the 2026 journal submission process.

Public Research References

Research Portal: <https://abhishekdemouser.github.io/vorgentia/>
Research Records: Figshare DOI, PhilArchive, PhilPapers, ORCID